

The Geodesic-Smoothness Question for Free Carnot Groups Has a Negative Literature Answer in General

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Status: answered negatively in the literature in general. Theorem 1.3 of arXiv:2603.23068v3 constructs non-smooth minimizing geodesics in a rank-two Carnot group. The standard free-group quotient lift transfers one to $G^9(\mathbb{R}^2)$, and hence to $G^m(\mathbb{R}^d)$ for every $m \geq 9$, $d \geq 2$. This refutes blanket smoothness beyond step two, but does not settle steps 3 through 8 individually.

1 The 2004 question

In Remark 14 of Friz and Victoir, *A note on the notion of geometric rough paths*, arXiv:math/0403115 [1], the authors would like to choose, for each $g \in G^m(V)$, a shortest Carnot–Caratheodory connection from the identity to g . They record smoothness of these minimizing geodesics as open for $m \geq 3$, contrasting it with the affirmative step-two theory.

By definition of the Carnot–Caratheodory norm, for every $\varepsilon > 0$, there exists a smooth path y , which we may take of constant speed (by time reparametrization), such that $S_m(y)_{0,1} = g$ and such that the sup norm of $\dot{y}$ is bounded by $\|g\| + \varepsilon$. Taking $\varepsilon = \|g\|$ finishes the first part. The second part follows from Proposition 11. ■

Remark 14 We would have liked to define h_g as a geodesic associated to g , e.g. the shortest connection of the neutral element in $G^m(V)$ with g w.r.t. the Carnot–Caratheodory distance $d_{G^m(V)}$. Unfortunately, smoothness of such geodesics is still an open problem for $m \geq 3$ (personal communication, R.Montgomery). An affirmative answer for the case $m = 2$ is found in [13].

Theorem 15 Let ω be a control satisfying condition (H_p) . A path Y belongs to $C^{\omega,p}(G^m(V))$ if and only if there exists a sequence of smooth V -valued paths $y(n)$ such that
(i): $\sup_n \|S_m(y(n))\| < \infty$.

Here $G^m(V)$ is the free step- m nilpotent (Carnot) group over the first layer V , the same free groups denoted below by $F_{d,m}$ when $V = \mathbb{R}^d$.

2 The 2026 negative theorem

Rossi, Schiavoni Piazza, and Socionovo prove the following in arXiv:2603.23068v3 [2].

Our next main result shows that the minimizing geodesics obtained in Theorem [1.1](#) can be lifted to minimizing geodesics in a Carnot group preserving both the regularity and the branching. This gives the first example of its kind.

Theorem 1.3. *There exist Carnot groups admitting minimizing geodesics that are non-smooth at an interior point of their domain and that exhibit branching.*

Carnot groups are a special class of sub-Riemannian manifolds, in the sense they are equiregular, self-similar and isometrically homogeneous. This theorem shows that neither equiregularity nor the group structure is enough to rule out non-smooth or branching geodesics.

Their Section 10 constructs the Carnot group explicitly. Its horizontal layer has the orthonormal frame Y_1, Y_2 , and a submetry lifts a minimizing geodesic that is non-smooth at an interior point while preserving its regularity and minimality. Thus the all-geodesics-smooth statement is false already within Carnot groups.

3 Transfer to the exact free groups

The resolving paper does not phrase Theorem 1.3 using rough-path notation, so we record the short standard implication.

Proposition 1 (Free lift of a non-smooth minimizer). *Let G be a rank- r , step- s Carnot group whose first layer has an orthonormal generating frame. If G admits a non-smooth minimizing geodesic, then the free Carnot group $F_{r,s} = G^s(\mathbb{R}^r)$ admits one. The same follows for every $F_{d,m}$ with $d \geq r$ and $m \geq s$.*

Proof. The universal property of $F_{r,s}$ gives a stratified epimorphism $\pi : F_{r,s} \rightarrow G$ mapping the free orthonormal generators to those of G . It is a metric submetry. Translate a non-smooth minimizing geodesic γ so that it starts at the identity and lift its horizontal control to a curve $\tilde{\gamma}$ starting at the identity of $F_{r,s}$. The lift preserves length. Therefore

$$d_G(e, \gamma(T)) \leq d_{F_{r,s}}(e, \tilde{\gamma}(T)) \leq L(\tilde{\gamma}) = L(\gamma) = d_G(e, \gamma(T)).$$

All inequalities are equalities, so $\tilde{\gamma}$ is minimizing. If it were smooth, then $\gamma = \pi \circ \tilde{\gamma}$ would be smooth, a contradiction. For larger rank or step, apply the same argument to the natural stratified epimorphism onto G . $\square$

In the lowest-regularity construction of [2], the parameter is $b = 5$. The Section 10 frame contains central-coordinate coefficients of degree at most $b + 3$, and the top coefficient requires $b + 4$ horizontal generators under iterated bracketing. Thus its Carnot group has step $s = b + 4 = 9$. The proposition gives a non-smooth minimizing geodesic in

$$G^9(\mathbb{R}^2) = F_{2,9},$$

and in every $G^m(\mathbb{R}^d)$ with $m \geq 9$ and $d \geq 2$.

4 What is and is not resolved

This is a decisive negative answer to any uniform claim that minimizing geodesics in all free groups $G^m(V)$ are smooth once $m \geq 3$. It also shows that the smooth step-two theory cannot extend without qualification.

Two narrower questions are not answered by this transfer. First, the individual free steps $3 \leq m \leq 8$ are not settled here. Second, a non-smooth minimizer between two endpoints does not exclude the possibility that the same endpoints also admit a different smooth minimizer. Therefore the result refutes universal smoothness of geodesics, but does not by itself exclude every possible rule selecting one smooth shortest connector per endpoint.

The source's Proposition 13 is unaffected: it deliberately uses a smooth connector whose length is within a factor two of optimal, rather than an exact minimizer.

References

- [1] P. Friz and N. Victoir, *A note on the notion of geometric rough paths*, arXiv:math/0403115 (2004).
- [2] T. Rossi, A. J. A. Schiavoni Piazza, and A. Socionovo, *Interior singularity and branching of geodesics in real-analytic sub-Riemannian manifolds*, arXiv:2603.23068v3 (2026).